Putnam

Counting and configurations

Written by a model for Adit. Not Adit's words.

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1 Counting and configurations

Recognise it. Arrangements, colourings, tilings, subsets with a property, lattice points, sequences with a forbidden pattern. People and acquaintances, teams and matches, committees and members, which are graphs and incidence structures in plain clothes. The question is how many, or whether a configuration exists, or whether it is impossible, or the largest size a configuration can have. The last two usually need two arguments: a proof that nothing bigger works and a construction that achieves the bound.

Theorem 1.1 (Double counting)**.** *Let A and B be finite sets and $R \subseteq A \times B$. For $a \in A$ let $d(a)$ be the number of b with $(a,b) \in R$, and for $b \in B$ let $d(b)$ be the number of a with $(a,b) \in R$. Then $\sum_{a \in A} d(a) = |R| = \sum_{b \in B} d(b)$.*

Proof. Both sides count the pairs in R, once grouped by the first coordinate and once by the second. □

Consequences used constantly: in a finite graph the degrees sum to twice the number of edges, so the number of odd-degree vertices is even; in a family of k-element subsets of an n-set in which every element lies in exactly r sets, $kF = rn$ where F is the number of sets; and $\sum_k k\binom{n}{k} = n2^{n-1}$, by counting pairs (subset, marked element) both ways.

Theorem 1.2 (Bijections and inclusion-exclusion)**.** *If $f\colon X \to Y$ is a bijection of finite sets then $|X| = |Y|$. For finite sets $A_1, \dots, A_n$ inside a finite U,*

$$\Big|\bigcup_i A_i\Big| = \sum_{\emptyset \neq S \subseteq \{1,\dots,n\}} (-1)^{|S|+1} \Big|\bigcap_{i \in S} A_i\Big|,$$

and the number of elements of U in none of the A_i is $|U|$ minus this, which for $n = 1$ is complementary counting.

Proof. For a bijection, pair each element with its image. For inclusion-exclusion, fix $u \in U$ and let m be the number of A_i containing it. Pointwise, $1 - \prod_i (1 - \mathbf{1}_{A_i}(u))$ equals 1 if $m \geq 1$ and 0 if $m = 0$; expanding the product gives $\sum_{S \neq \emptyset} (-1)^{|S|+1} \prod_{i \in S} \mathbf{1}_{A_i}(u)$. Sum over u. □

Theorem 1.3 (Averaging and the probabilistic method)**.** *If a finite set of objects has total score T and there are N objects, some object has score at least T/N and some object has score at most T/N. If an object is chosen at random and the expected number of "bad events" is less than 1, some object has no bad event.*

Proof. If every score were less than T/N the total would be less than T. The second statement is the first applied to the number of bad events, which is a non-negative integer, so an object with fewer than 1 of them has none. □

Proposition 1.4 (Colouring arguments for tilings)**.** *Suppose the cells of a board are coloured, and every tile of a given shape, however placed, covers cells whose colour counts form the same vector $t = (t_1, \dots, t_c)$, or more generally whose colour counts satisfy a fixed linear relation. If the board's colour counts $(b_1, \dots, b_c)$ are not a non-negative integer multiple of t (or fail the relation), the board cannot be tiled by that shape. The argument proves only impossibility; a possible case still needs an explicit tiling.*

Proof. A tiling with N tiles covers each cell once, so the colour counts of the board equal the sum over tiles of their colour counts, which is Nt. □

The standard colourings: the chessboard colouring, under which a domino covers one cell of each colour; the colouring of column j by j mod k, under which a $1 \times k$ bar covers every colour once but a $k \times 1$ bar covers k cells of one colour; and a colouring by $(i + j)$ mod k or $(i + 2j)$ mod k, chosen so that each placement of an awkward tile covers a prescribed pattern. Weighted colourings, with a number in each cell, are the same argument with the sum in place of the count.

Lemma 1.5 (Cycles)**.** *A finite graph in which every vertex has degree at least 2 contains a cycle. A connected graph on n vertices has at least $n - 1$ edges, with equality exactly for trees, and any graph on n vertices with at least n edges contains a cycle.*

Proof. Walk from any vertex, never immediately reversing; degree at least 2 makes this always possible, and the finite graph forces a repeated vertex, which closes a cycle. For the second part, remove edges from cycles until none remain; connectivity is preserved and a connected acyclic graph has exactly $n-1$ edges (induction on n, removing a vertex of degree 1, which exists by the first statement since there is no cycle and the graph is connected). □

First moves. List every object for $n = 2$ and $n = 3$ and count by hand. Colour the board. Count by rows and by columns, or by objects and by properties, and compare. Translate people and relations into a graph and ask about degrees. Build a bijection to something already counted: binary strings, lattice paths, subsets. For a maximum, guess the extremal configuration from the small cases, prove the bound by double counting or averaging, and present the construction separately. For a parity claim, look for an involution without fixed points.

Example 1.6. For a nonempty subset S of $\{1, \dots, n\}$, let $w(S) = \max S - \min S$. Find $\sum_S w(S)$ over all nonempty subsets S.

Solution. Count pairs (S, k) with S nonempty and $\min S \le k < \max S$, where k is an integer. Grouped by S, there are $w(S)$ such k for each S, so the number of pairs is the sum we want. Grouped by k, with $1 \le k \le n - 1$, the subsets S that qualify are

those containing at least one element in $\{1,\dots,k\}$ and at least one in $\{k+1,\dots,n\}$; there are $(2^k-1)(2^{n-k}-1)$ of them. Hence

$$\sum_S w(S) = \sum_{k=1}^{n-1}(2^k-1)(2^{n-k}-1) = \sum_{k=1}^{n-1}\big(2^n-2^k-2^{n-k}+1\big) = (n-1)2^n - 2(2^n-2) + (n-1),$$

which simplifies to $(n-3)2^n+n+3$.

Check at $n=2$: only $\{1,2\}$ contributes, with $w=1$; the formula gives $-4+5=1$. At $n=3$: the three pairs contribute $1,2,1$ and the full set contributes 2, total 6; the formula gives $0+6=6$. The sum could also be done as $\sum_S \max S - \sum_S \min S$ with $\sum_S \max S = \sum_k k2^{k-1}$, but the pair count did both at once and needed no identity for $\sum k2^{k-1}$. □

Example 1.7. Let $n \geq 1$. Find the number of pairs (A,B) of subsets of $\{1,\dots,n\}$ with $A \subseteq B$ and $|B \setminus A|$ odd.

Solution. Each element is in one of three states: outside B, in $B\setminus A$, or in A. So there are 3^n pairs in all, and the number with exactly k elements in $B\setminus A$ is $\binom{n}{k}2^{n-k}$. The answer is $\sum_{k \text{ odd}}\binom{n}{k}2^{n-k}$.

Rather than sum this, pair the configurations off. Given (A,B) with $A \neq \{1,\dots,n\}$, let j be the smallest element not in A and move it between the states "outside B" and "in $B\setminus A$". This changes $|B\setminus A|$ by one, so it swaps odd and even, and doing it twice returns to the start; it is an involution on the pairs with $A \neq \{1,\dots,n\}$ with no fixed points. The only unpaired configuration is $A=B=\{1,\dots,n\}$, with $|B\setminus A|=0$, even. So the number of even pairs exceeds the number of odd pairs by exactly 1, and with 3^n pairs in total the number of odd pairs is $(3^n-1)/2$.

Check at $n=1$: the pairs are $(\emptyset,\emptyset)$, $(\emptyset,\{1\})$, $(\{1\},\{1\})$, and exactly one has an odd difference; $(3-1)/2=1$. At $n=2$: nine pairs, and the odd ones have $|B\setminus A|=1$, of which there are $\binom{2}{1}\cdot 2 = 4 = (9-1)/2$. The binomial sum confirms it: $\sum_{k \text{ odd}}\binom{n}{k}2^{n-k} = \frac{1}{2}\big((2+1)^n-(2-1)^n\big)$, which is the generating-function view of the same involution. □

Trap. Overcounting symmetric arrangements: a necklace counted as a sequence, a handshake counted from both ends, a pair counted as an ordered pair. A bijection that is not onto, or not well defined on the whole domain, which is why the inverse map should be written down. A colouring that proves impossibility and forgets the construction for the possible cases, or a construction given for one n when the claim is for all n. Inclusion-exclusion with the signs reversed or a term missed; write the $n=2$ case to fix the pattern. Reading a graph problem as being about the specific people in it rather than the degrees. A maximum claimed from the small cases without the proof that nothing larger exists, or proved without the example that attains it.

Ladder.

1. PS 10 #1. A square grid and an L-shaped tile; can it be tiled.
2. PS 10 #4. Digit strings with a divisibility condition; how many.
3. PS 10 #7. The first n integers into boxes of equal sum; when possible.
4. 2018 B1. A set of lattice vectors; remove one so the rest split into two equal-sum halves.
5. 2019 B1. Lattice points on a family of circles; count the squares among them.
6. 2023 B1. Coins sliding on a grid; which configurations are reachable.
7. 2024 B1. A grid with prescribed values along a transversal; feasibility.

8. 2017 A4. Quiz scores; split the class into two halves with equal averages.
9. 2022 B4. Cyclic sequences in which consecutive triples form progressions; which lengths.
10. 2016 A4. Tiling a rectangle with a given tile; the minimum number of tiles.
11. 2025 A5. Permutations with a sign pattern of differences; the pattern maximising the count.
12. 2024 A3. Order statistics of a random arrangement in a rectangle; a balanced pair exists.

Where it hides. Everywhere. A number theory problem that asks how many residues or how many integers of a form (2020 A1, 2022 A3). A probability that is a count divided by a total (2021 B1, 2024 A3). A linear algebra problem whose matrix is an incidence matrix (2018 A2, 2022 B2). A geometry problem about lattice points (2019 B1, 2025 B1). An invariant problem where the reachable set has to be counted once the invariant is found (2023 B1, 2024 B1). A recurrence whose terms count something (2020 A5, 2024 B5).